

A Point-Like Dark-Matter Target at 12.8 GHz from a QCD-Anchored Vacuum Condensate: Theory and Haloscope Test Protocol

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August 5, 2026

Abstract

The Null-Vector Gravity (NVG) framework, anchored to a single QCD input $M_{\Omega,0} = 859 \pm 8$ MeV fixed by lattice sigma terms, contains a pseudo-Goldstone mode θ of the vacuum condensate $\Phi = W e^{i\theta}$. We examine the phenomenological consequences of the hypothesis that the same decay constant f_a that sets the θ -mode mass also generates Majorana neutrino masses through a Weinberg-like dimension-five operator suppressed by a two-loop QCD-anomaly coefficient. Introducing the unknown overall coefficient of this operator as an explicit parameter c_ν , the benchmark choice $c_\nu = 1$ fixes $f_a = 1.07 \times 10^{11}$ GeV from the neutrino sector and yields a *conditional point target* for dark matter: a coherent pseudo-scalar of mass $m_\theta = 53.2$ μ eV, detectable as a microwave line at $f_\theta = 12.86$ GHz with photon coupling $g_{\theta\gamma\gamma} = 2.08 \times 10^{-14}$ GeV $^{-1}$ (KSVZ benchmark). The product $m_\theta f_a$ agrees with the QCD axion band at 2.6% — a normalization cross-check, not an independent test — and the benchmark frequency lies in the currently unexplored gap between the published ORGAN mass windows (25–26 and 63–67 μ eV), directly on the lower edge of the SMASH band (50–200 μ eV) that ORGAN has declared as its roadmap target. We present the derivation, a calibration of the haloscope signal formula against the published ORGAN signal estimate, a Monte Carlo propagation of the statistical input uncertainties (m_3, χ_{top}), together with explicit conditional sensitivity scenarios for the unknown coefficient c_ν , and two concrete test protocols: (A) a resonant-cavity array and (B) a *fixed-position* dielectric booster stack, where the point target eliminates the scanning mechanism. For a coherent dark-matter fraction of 24% (misalignment only), protocol B reaches SNR = 5 in ~ 18 h at power boost $\beta^2 = 10^4$ (MADMAX convention, $\beta^2 \equiv P_{\text{sig}}/P_0$). The target is extended to nucleon, electron and neutrino channels with no additional free parameters, and Appendix 8 gives a perturbative UV consistency example for the θ -seesaw (it does not derive the QCD-anomaly suppression). Discovery and exclusion criteria are preregistered. This is a *phenomenological target paper*: the θ -seesaw operator is stated as a hypothesis, its coefficient is propagated as a parameter, and every derived quantity is conditional on it. All numerical values are reproducible from the accompanying scripts.

Keywords: axion dark matter, haloscope, dielectric haloscope, pseudo-Goldstone boson, vacuum condensate, neutrino mass, phenomenological target

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1 Introduction

Direct searches for QCD axion dark matter have so far produced exclusion limits in disjoint mass windows: ADMX covers $\sim 2.7\text{--}4.2\ \mu\text{eV}$ [14], HAYSTAC covers $\sim 17\text{--}20\ \mu\text{eV}$ [15], ORGAN has scanned $25.45\text{--}26.27\ \mu\text{eV}$ [18], $63.2\text{--}67.1\ \mu\text{eV}$ [16] and $107.42\text{--}111.93\ \mu\text{eV}$ [17], while MADMAX is under construction for the $40\text{--}400\ \mu\text{eV}$ range [21]. The intermediate region around $50\text{--}60\ \mu\text{eV}$ ($12\text{--}15\ \text{GHz}$) remains unexplored.

This note reports that the NVG/VMF framework [2], anchored to a single QCD input [1], yields — conditional on one stated hypothesis (the θ -seesaw, Sec. 2.3) — a point target *exactly* in that gap: a dark-matter pseudo-scalar at $53.2\ \mu\text{eV}$. Because the target is a *point* in (m, g_γ) space rather than a band, the experimental problem simplifies drastically: no scanning, no tuning mechanism — a fixed-frequency integration.

Section 2 gives the derivation. Section 3 states the observable signature. Section 5 documents the gap in published searches. Section 6 defines the test protocol, including a calibration of the signal formula against published data and preregistered discovery/exclusion criteria. Section 4 extends the target to nucleon, electron and neutrino channels; Appendix 8 gives a perturbative UV consistency example for the θ -seesaw. Section 7 lists the framework assumptions honestly.

2 Theoretical Framework

2.1 The anchor: $M_{\Omega,0}$ from lattice sigma terms

The framework is anchored to one dimensionful QCD input. Separating the nucleon mass into current-quark and nonperturbative components via the Feynman–Hellmann sigma terms

($\sigma_{\pi N} \approx 44$ MeV, $\sigma_{sN} \approx 30$ MeV), the vacuum value of the nonperturbative component is fixed at [1]

$$M_{\Omega,0} = 859 \pm 8 \text{ MeV}, \quad (1)$$

which sets the condensate energy density scale $\rho_c \approx 7.09 \times 10^4$ MeV/fm³ used throughout the construction. No cosmological or astrophysical input is used at any stage.

2.2 The θ pseudo-Goldstone mode

The NVG vacuum condensate is parametrized as $\Phi = W e^{i\theta}$, where the phase θ is a pseudo-Goldstone mode with the standard anomaly-generated potential. Its mass obeys the same relation as the QCD axion,

$$m_\theta = \frac{\sqrt{\chi_{\text{top}}}}{f_a}, \quad (2)$$

with the topological susceptibility $\chi_{\text{top}}^{1/4} = 75.5$ MeV from lattice QCD [10]. In the normalization $m_a(f_a = 10^{12} \text{ GeV}) = 5.691 \text{ } \mu\text{eV}$ [10], this becomes $m_\theta = 5.691 \text{ } \mu\text{eV} \times (10^{12} \text{ GeV}/f_a)$.

2.3 The θ -seesaw: fixing f_a from the neutrino sector

The framework hypothesis behind the target is the θ -seesaw: the same mode θ generates Majorana neutrino masses without right-handed neutrinos. To be consistent with the electroweak gauge group above EWSB, the operator must have the unique gauge-invariant dimension-five (Weinberg) structure [7]; the NVG hypothesis specifies only its suppression scale and coefficient,

$$\mathcal{L}_{\text{eff}} = 2 c_\nu \left(\frac{\alpha_s}{4\pi} \right)^2 \frac{1}{f_a} (\tilde{H}^\dagger L)^T C (\tilde{H}^\dagger L) + \text{h.c.}, \quad \tilde{H} \equiv i\sigma_2 H^*, \quad (3)$$

where c_ν is an unknown dimensionless coefficient. The normalization convention of this Weinberg-like operator, including symmetry factors in flavor space and the factors of $1/\sqrt{2}$ from $\langle H \rangle = v_{\text{EW}}/\sqrt{2}$, is absorbed into c_ν ; accordingly, $c_\nu = 1$ defines a benchmark normalization rather than a UV-derived coefficient. The operator is *defined* through the physical post-EWSB relation, for the heaviest neutrino,

$$m_3 = c_\nu \left(\frac{\alpha_s}{4\pi} \right)^2 \frac{v_{\text{EW}}^2}{f_a}. \quad (4)$$

With $\alpha_s(M_Z) = 0.1184$, $v_{\text{EW}} = 246.22$ GeV and the oscillation-and-cosmology anchor $m_3 = 50.3$ meV (normal ordering; the implied sum $\sum m_\nu = 59$ meV passes the DESI DR2 bound $\sum m_\nu < 64$ meV unconditionally), Eq. (4) fixes

$$\boxed{f_a = c_\nu \left(\frac{\alpha_s}{4\pi} \right)^2 \frac{v_{\text{EW}}^2}{m_3} = 1.07 \times 10^{11} c_\nu \text{ GeV}} \quad (5)$$

We stress the status of Eq. (3): it is a *framework hypothesis*, the single non-standard ingredient of this note. The nominal coefficient $(\alpha_s/4\pi)^2 \approx 8.9 \times 10^{-5}$ is motivated by the two-loop QCD-anomaly diagram, but possible flavor factors and RG running are not computed here, and the UV completion that generates the operator is not derived. The parameter c_ν absorbs all of this freedom: the benchmark $c_\nu = 1$ defines the headline target, and every result below is rescaled with c_ν (Section 6.4). The haloscope test discriminates the hypothesis directly (Section 7).

2.4 Point target for the dark-matter mass

Because every derived quantity depends on c_ν through f_a alone, the parametric structure is transparent (for fixed E/N):

$$f_a = f_{a,0} c_\nu, \quad m_\theta = \frac{m_{\theta,0}}{c_\nu}, \quad f_\theta = \frac{f_{\theta,0}}{c_\nu}, \quad g_{\theta\gamma\gamma} = \frac{g_{\theta\gamma\gamma,0}}{c_\nu}, \quad (6)$$

where the subscript 0 denotes the benchmark values at $c_\nu = 1$. A larger c_ν means a larger f_a , hence a *smaller* mass and frequency and a weaker photon coupling. Inserting (5) into (2):

$$m_\theta = 5.691 \text{ } \mu\text{eV} \times \frac{10^{12}}{1.07 \times 10^{11} c_\nu} = \frac{53.2}{c_\nu} \text{ } \mu\text{eV} \quad (7)$$

or, equivalently, a Compton frequency

$$f_\theta = \frac{m_\theta c^2}{h} = \frac{12.86}{c_\nu} \text{ GHz}. \quad (8)$$

The benchmark $c_\nu = 1$ gives $m_\theta = 53.2 \text{ } \mu\text{eV}$, $f_\theta = 12.86 \text{ GHz}$ throughout the rest of this note.

Point versus band. If a UV completion fixes c_ν , the target is a genuine *conditional point* in (m, g_γ, f) space. If c_ν remains free, the point degrades to a *broad conditional search band* along the one-parameter curve (6) — 4.3–38.6 GHz for $c_\nu \in [1/3, 3]$ (Sec. 6.4) — and the fixed-frequency advantage is retained only conditionally, at a chosen benchmark value of c_ν . We use the terminology “conditional point target” and “broad conditional search band” accordingly.

Normalization cross-check (not an independent test). Once Eq. (2) is accepted, the product $m_\theta f_a = \sqrt{\chi_{\text{top}}}$ is fixed by the same QCD input, so its landing on the QCD axion band $m_a f_a \simeq m_\pi f_\pi \sqrt{z}/(1+z) \approx 5.82 \times 10^{15} \text{ eV}^2$ [10] cannot count as an independent confirmation of NVG or of the seesaw hypothesis. The framework gives $m_\theta f_a = 5.67 \times 10^{15} \text{ eV}^2$: the 2.6% deviation is a numerical consistency check of two lattice normalizations (the χ_{top} of [10] against the VMF anchor), and nothing more. The genuinely independent content of the target is the *value* of f_a itself, fixed by the neutrino sector.

2.5 Coupling to photons

The effective interaction is the standard axion-electrodynamics term [8]

$$\mathcal{L} \supset -\frac{g_{\theta\gamma\gamma}}{4} \theta F_{\mu\nu} \tilde{F}^{\mu\nu} = g_{\theta\gamma\gamma} \theta \mathbf{E} \cdot \mathbf{B}, \quad g_{\theta\gamma\gamma} = \frac{\alpha}{2\pi} \frac{|E/N - 1.92|}{f_a}. \quad (9)$$

For the KSVZ benchmark [5] ($E/N = 0$) and the DFSZ benchmark [6] ($E/N = 8/3$):

$$g_{\theta\gamma\gamma}^{\text{KSVZ}} = 2.08 \times 10^{-14} \text{ GeV}^{-1}, \quad g_{\theta\gamma\gamma}^{\text{DFSZ}} = 8.10 \times 10^{-15} \text{ GeV}^{-1}. \quad (10)$$

The model-dependent factor $|E/N - 1.92|$ is the only residual theory uncertainty on the coupling. We emphasize that $g_{\theta\gamma\gamma}$ is *not predicted* by NVG: the framework fixes f_a but not the UV anomaly ratio E/N , so KSVZ and DFSZ are benchmarks, not consequences of the model. A null result at KSVZ level therefore excludes only KSVZ-like realizations (Section 6.4). The KSVZ value is used as the discovery benchmark.

2.6 Relic abundance: the honest coherent fraction

For a post-inflationary scenario with the natural initial misalignment $\theta_i = \pi/\sqrt{3}$, the standard misalignment estimate [9] gives

$$\Omega_\theta h^2 \simeq 0.12 \theta_i^2 \left(\frac{f_a}{10^{12} \text{ GeV}} \right)^{7/6} = 0.0291, \quad (11)$$

i.e. **24% of the local dark-matter density** ($\Omega_{\text{DM}} h^2 = 0.120$). The remaining density may reside in topological defects of the condensate sector; it does not contribute to the coherent haloscope signal. All signal powers below use the conservative coherent fraction $f_{\text{coh}} = 0.24$; the $f_{\text{coh}} = 1$ values are also quoted where relevant. A larger coherent fraction (e.g. from defect decay) would strengthen the signal proportionally.

2.7 Fifth force: the pseudoscalar null channel

The Compton range $\lambda_\theta = \hbar c/m_\theta = 3.72$ mm falls inside the sub-millimetre window of torsion-balance experiments [12]. Because θ is a *pseudoscalar*, the leading-order monopole–monopole Yukawa exchange between unmixed states vanishes, so unpolarized Eöt-Wash-class searches are expected to see nothing at this range. The residual channels — spin-dipole exchange with coupling m_N/f_a , loop-induced scalar admixtures, two-boson exchange, or condensate mixing — require a dedicated analysis and are not claimed here as quantitative predictions; we state only the robust null expectation for the unpolarized monopole channel.

3 Experimental Signature

The benchmark target is summarized in Table 1.

Table 1: Benchmark target of the NVG θ -DM sector, conditional on the θ -seesaw hypothesis with $c_\nu = 1$. No quantity in the right column is fitted to haloscope data.

Quantity	Predicted value	Origin
Decay constant f_a	1.07×10^{11} GeV	neutrino sector, Eq. (4)
Mass m_θ	53.2 μ eV	Eq. (2), lattice χ_{top}
Frequency f_θ	12.86 GHz	$m_\theta c^2/h$
Virial linewidth Q_a	$\sim 10^6$	halo velocity dispersion
Line width Δf_a	12.8 kHz	f_θ/Q_a
$g_{\theta\gamma\gamma}$ (KSVZ)	2.08×10^{-14} GeV $^{-1}$	Eq. (9)
$g_{\theta\gamma\gamma}$ (DFSZ)	8.10×10^{-15} GeV $^{-1}$	Eq. (9)
Coherent DM fraction	24%	misalignment, Sec. 2.6
$m_\theta f_a$ vs QCD band	−2.6% deviation	consistency check

The signal is a nearly monochromatic microwave excess at 12.86 GHz with $Q_a \sim 10^6$ intrinsic quality factor, linearly polarized along the applied magnetic field, scaling as B_0^2 , and tracking the sidereal modulation of the halo wind (a secondary discriminant).

4 Multi-Channel Interaction Block

The photon coupling of Sec. 2 is one entry of the full low-energy interaction block. Completing it makes the target multi-channel falsifiable without introducing new free parameters: the fermion axial charges below are fixed either by the same UV benchmark choice (KSVZ-like or DFSZ-like) already made for E/N , or — in the neutrino channel — by the θ -seesaw operator itself.

4.1 Effective Lagrangian and channel benchmarks

$$\mathcal{L}_{\text{int}} = -\frac{g_{\theta\gamma\gamma}}{4} \theta F \tilde{F} + \frac{\partial_\mu \theta}{2f_a} \sum_f C_f \bar{f} \gamma^\mu \gamma^5 f, \quad (12)$$

with $f = e, p, n, \nu_3$. Using the fermion equations of motion, each derivative coupling is equivalent to a pseudoscalar Yukawa coupling $g_{\theta ff} = C_f m_f/f_a$.

Nucleons. The axial charges split into a model-independent chiral piece (axion–pion mixing, fixed by QCD) and the UV charges C_q^0 [27]:

$$C_p = -0.47 + 0.88 C_u^0 - 0.39 C_d^0 - 0.49 C_s^0, \quad C_n = -0.02 - 0.02 C_u^0 + 0.88 C_d^0 - 0.49 C_s^0. \quad (13)$$

For the KSVZ-like benchmark ($C_q^0 = 0$ — the same UV choice as $E/N = 0$), $C_p = -0.47$ and $C_n = -0.02$ are *predictions*, not parameters: $g_{\theta pp} = -4.1 \times 10^{-12}$ and $g_{\theta nn} = -1.8 \times 10^{-13}$. The spin-dependent fifth-force channel of Sec. 3 is thereby upgraded from the order-of-magnitude quote m_N/f_a to a definite coupling.

Electrons. The two photon benchmarks separate here: DFSZ-like UV gives a tree-level $C_e = \cos^2 \beta / 3$ (illustratively 0.033 at $\tan \beta = 3$, $g_{\theta ee} \simeq 1.6 \times 10^{-16}$; smaller $\tan \beta$ raises C_e up to $1/3$), while KSVZ-like UV gives $C_e = 0$ at tree level and a loop-suppressed $C_e \sim 10^{-5}$ ($g_{\theta ee} \sim 5 \times 10^{-20}$). The resulting ~ 3 –4-order spread makes the electron channel a UV discriminator. Both values pass the white-dwarf cooling constraint $g_{ae} \lesssim 3 \times 10^{-13}$ [12] with wide margins.

Neutrinos. The θ -seesaw operator itself carries the PQ phase, so the heaviest-state mass term reads $m_3 e^{i\theta/f_a}$: the heaviest neutrino obtains an axial charge $C_{\nu_3} = 1$ in the normalization of Eq. (3), i.e. $g_{\theta \nu_3 \nu_3} \simeq m_3/f_a = 4.7 \times 10^{-22}$. This coupling is a distinctive NVG signature — it is absent in plain KSVZ and DFSZ — and the decay $\theta \rightarrow \nu \nu$ is kinematically forbidden ($2m_3 \gg m_\theta$), so the channel affects stability of the dark-matter candidate not at all. The solar mass m_2 , generated by a second operator sector without PQ charge (Sec. 4.2), carries no such coupling.

Table 2: Multi-channel benchmark couplings at $f_a = 1.07 \times 10^{11}$ GeV ($c_\nu = 1$), all scaling as $1/c_\nu$. Rows marked “benchmark” inherit the same UV choice as the photon channel; the neutrino row and the flavor benchmark of Sec. 4.2 follow from the θ -seesaw itself. The DFSZ-like column is an illustrative UV benchmark ($\tan \beta = 3$ where a value is shown).

Channel	KSVZ-like benchmark	DFSZ-like (illustrative)
Photon $g_{\theta \gamma \gamma}$ (GeV $^{-1}$)	2.1×10^{-14}	8.1×10^{-15}
Nucleon $g_{\theta pp}$	-4.1×10^{-12}	$\sim -6 \times 10^{-12}$ ($\tan \beta = 3$)
Nucleon $g_{\theta nn}$	-1.8×10^{-13}	$\sim +8 \times 10^{-14}$ ($\tan \beta = 3$)
Electron $g_{\theta ee}$	$\sim 5 \times 10^{-20}$ (loop)	1.6×10^{-16} (tree, $\tan \beta = 3$)
Neutrino $g_{\theta \nu_3 \nu_3}$	4.7×10^{-22} (NVG-specific)	

4.2 Minimal normal-ordering flavor benchmark

The two measured mass splittings require at least two massive light neutrinos. Accordingly, the seesaw sector contains at least two heavy states: a PQ-carrying state N_3 that generates the dominant atmospheric mass m_3 (and hence fixes f_a and the haloscope benchmark via Eq. (4)), and a PQ-neutral state N_2 that generates the solar mass m_2 and does not couple to θ at leading order. With $m_1 \simeq 0$, this *minimal normal-ordering benchmark* and the NuFIT 5.2 oscillation parameters [30] yield the correlated triple

$$\sum m_\nu = 58.9 \text{ meV}, \quad m_\beta = 8.8 \text{ meV}, \quad m_{\beta\beta} \in [1.5, 3.7] \text{ meV}, \quad (14)$$

where the $0\nu\beta\beta$ value carries a *nonzero floor* ($|U_{e2}^2 m_2 - U_{e3}^2 m_3|$ at destructive Majorana phase). The inverted-ordering alternative gives $\sum m_\nu \approx 106$ meV, so the minimal-NO hypothesis is a clean cosmological discriminator: it places the mass sum at the normal-ordering minimum, and a measurement $\sum m_\nu \gtrsim 75$ meV (CMB-S4/DESI-class sensitivity) would exclude it. As with the c_ν scenarios of Sec. 6.4, the minimal spectrum and the N_2 sector are scenario assumptions, not framework derivations.

4.3 Experimental mapping

The predicted nucleon couplings fall inside the design window of NMR/spin-precession programs: the CASPER design band $f_a \sim 10^9\text{--}10^{12}$ GeV [29] contains $f_a = 1.07 \times 10^{11}$ GeV, and at $m_\theta = 53.2 \mu\text{eV}$ the oscillating nuclear EDM channel (CASPER-electric type) and the spin-gradient channel (CASPER-wind, QUAX, ARIADNE type) address the same mass with a *predicted* coupling rather than a scanned one. The electron channel is a UV discriminator rather than a prediction; the fifth-force channel of Sec. 3 remains qualitative pending a computation of the loop-induced scalar admixture.

5 Status of Published Searches

Table 3 compiles the published haloscope coverage around the predicted mass.

Table 3: Published haloscope mass windows around the NVG target. The benchmark point ($53.2 \mu\text{eV}$) lies in the unexplored gap between the two ORGAN windows, at the lower edge of the SMASH band ($50\text{--}200 \mu\text{eV}$) [11] that ORGAN declares as its roadmap target [16].

Experiment	Mass window (μeV)	Frequency (GHz)	Best limit on $g_{a\gamma\gamma}$ (GeV^{-1})
ADMX [14]	2.66–3.31	0.64–0.80	KSVZ reached
HAYSTAC II [15]	$\sim 17.3\text{--}19.5$	4.18–4.71	$\sim 10^{-13}$
ORGAN [18]	25.45–26.27	6.15–6.35	2.8×10^{-13}
NVG target	53.2	12.86	2.08×10^{-14}
ORGAN [16]	63.2–67.1	15.3–16.2	3.0×10^{-12}
ORGAN 1b [17]	107.42–111.93	26.0–27.1	cogenesis ALP excluded
MADMAX [21, 24]	40–400 (design)	10–100	prototype phase

Two remarks. First, no published limit reaches $g = 2.1 \times 10^{-14} \text{ GeV}^{-1}$ anywhere near $53 \mu\text{eV}$: the closest result (ORGAN at $25.5\text{--}26.3 \mu\text{eV}$) is a factor ~ 13 weaker in coupling, at a different mass. Second, the ORGAN collaboration’s stated goal is to scan the SMASH band $50\text{--}200 \mu\text{eV}$ under a constant magnetic field [16]; the NVG benchmark sits exactly at the lower edge of that band, making a targeted single-frequency run a natural addition to their program.

6 Test Protocol

6.1 Signal-power formula and its calibration

For a microwave cavity haloscope [8, 25],

$$P_{\text{sig}} = g_{\theta\gamma\gamma}^2 \frac{\rho_\theta}{m_\theta} B_0^2 V C \min(Q_L, Q_a), \quad (15)$$

where $\rho_\theta = f_{\text{coh}} \rho_{\text{DM}}$ with $\rho_{\text{DM}} = 0.4 \text{ GeV}/\text{cm}^3$, V is the cavity volume, C the mode form factor, and Q_L the loaded quality factor.

Calibration against published data. The ORGAN collaboration quotes, for its Phase 1a apparatus, $P_{\text{sig}} \simeq 2.3 \times 10^{-25} \text{ W}$ (KSVZ) and $3.1 \times 10^{-26} \text{ W}$ (DFSZ) for a $64 \mu\text{eV}$ axion [16]. Two checks:

1. *Apparatus-independent*: the coupling-coefficient ratio $(g_{\text{KSVZ}}/g_{\text{DFSZ}})^2 = (0.97/0.36)^2 = 7.26$ versus the published power ratio $2.3/0.31 = 7.42$ — agreement at the 2% level, confirming the normalization convention of Eq. (9).

2. *Absolute*: back-solving Eq. (15) from the published 2.3×10^{-25} W implies an apparatus product $VCQ \approx 0.44 \text{ m}^3$ (for $B_0 = 7 \text{ T}$), consistent with an ORGAN-scale cavity at 15.6 GHz with $Q \sim 10^4$ – 10^5 . The unit chain of the present calculation is thus validated against a real haloscope at the factor-of-two level.

For the dielectric booster concept [19, 20, 21] the analogous expression is

$$P_{\text{sig}} = g_{\theta\gamma\gamma}^2 \rho_\theta B_0^2 \frac{A \beta^2}{m_\theta^2}, \quad (16)$$

where A is the disk area and $\beta^2 \equiv P_{\text{sig}}/P_0$ is the *power* boost relative to a bare magnetized mirror ($P_0 = g_{\theta\gamma\gamma}^2 \rho_\theta B_0^2 A/m_\theta^2$), following the MADMAX convention [24] in which β itself denotes the amplitude boost; all boost figures quoted in this note are power boosts β^2 . Both Eq. (15) and (16) have been re-derived within the Lorentz-reciprocity framework [25, 26]; an in-situ reciprocity measurement with a source antenna is the final normalization check of the stack before any dark-matter run.

6.2 Protocol A: resonant cavity

At $f_\theta = 12.86 \text{ GHz}$ the TM_{010} mode requires a cylinder of radius $R = x_{01}c/(2\pi f_\theta) = 8.92 \text{ mm}$. With length $L = 50 \text{ mm}$, $V = 12.5 \text{ cm}^3$, form factor $C = 0.69$, $B_0 = 10 \text{ T}$ and a superconducting (Nb) cavity with baseline $Q_L = 10^5$,

$$P_{\text{sig}} = 6.3 \times 10^{-25} \text{ W} \quad (f_{\text{coh}} = 0.24). \quad (17)$$

A single cavity reaches $\text{SNR} = 5$ only after ~ 48 years: it is a pathfinder, not the discovery configuration. The viable cavity route is a phase-locked array (Table 4); $Q_L = 10^6$ at 13 GHz is flagged optimistic (Nb BCS losses scale as ω^2), so the $Q = 10^5$ rows are the engineering baseline.

Table 4: Integration time to $\text{SNR} = 5$ for cavity arrays ($B_0 = 10 \text{ T}$, $T_{\text{sys}} = 1 \text{ K}$, $f_{\text{coh}} = 0.24$, per-cavity signal from Eq. (15)).

	$N = 64$	$N = 256$
$Q_L = 10^5$ (baseline)	276 days	69 days
$Q_L = 10^6$ (optimistic)	0.3 days	0.1 days

6.3 Protocol B: fixed-position dielectric booster stack

The dielectric haloscope [19] replaces the resonant cavity by ~ 80 dielectric disks (LaAlO_3 , $\varepsilon \approx 24$) in a strong magnetic field; constructive interference of the emitted photons provides the power boost β^2 . In a scanning experiment the disk positions must be moved mechanically — the technically hardest subsystem, now being commissioned [23]. **The point-frequency target eliminates this subsystem entirely**: at a known frequency the inter-disk spacing is fixed, and the stack becomes a static optical element.

With $D = 1.2 \text{ m}$ disks ($A = 1.13 \text{ m}^2$, the low-mass-end MADMAX aperture), $B_0 = 10 \text{ T}$ and $f_{\text{coh}} = 0.24$, Eq. (16) gives (Table 5). The power-boost targets $\beta^2 = 10^4$ – 10^5 are the design goals of the MADMAX program [21]; the three-disk prototype has so far demonstrated β^2 up to ~ 2000 with 13–17% uncertainty [24], which we state as the principal technical risk of this option.

Protocol B is strictly simpler than the cavity array: no cryogenic tuning mechanism, no phase-locked multi-cavity readout, a single wide-bore magnet, horn antenna and a near-quantum-limited amplifier — while the required power boost β^2 is already the stated MADMAX design target in a frequency band that includes 12.86 GHz.

Table 5: Fixed dielectric stack, $B_0 = 10$ T, $T_{\text{sys}} = 1$ K, readout bandwidth equal to the axion linewidth $\Delta f_a = 12.8$ kHz (the stack boost band $\gg \Delta f_a$).

Power boost β^2	$P_{\text{sig}} (f_{\text{coh}} = 0.24)$	$P_{\text{sig}} (f_{\text{coh}} = 1)$	$t(\text{SNR} = 5)$
10^4	3.1×10^{-23} W	1.3×10^{-22} W	18 h
10^5	3.1×10^{-22} W	1.3×10^{-21} W	11 min

6.4 Readout, statistics, and preregistered criteria

Noise. $T_{\text{sys}} \approx 1$ K is adopted: millikelvin physical temperature plus a flux-driven Josephson parametric amplifier. This is not an assumption but demonstrated technology — ORGAN’s most recent phase operates exactly this way [18].

Statistical target definition. With the Dicke radiometer relation

$$\text{SNR} = \frac{P_{\text{sig}}}{k_B T_{\text{sys}}} \sqrt{\frac{t}{\Delta f}}, \quad (18)$$

the narrow search window below is conditional on the benchmark coefficient $c_\nu = 1$. Statistical uncertainties in m_3 and χ_{top} are propagated by Monte Carlo (Appendix 8: $m_3 \sim \mathcal{N}(50.3, 0.25)$ meV, $\chi_{\text{top}}^{1/4} \sim \mathcal{N}(75.5, 0.7\%)$ MeV), while c_ν is *not* assigned a statistical prior: it is an explicit theory parameter. Broader intervals obtained by varying c_ν below are sensitivity scenarios, not confidence intervals.

- **Conditional benchmark window ($c_\nu = 1$ fixed).** The target-frequency distribution has median 12.86 GHz and a 68% interval [12.67, 13.05] GHz (relative width 1.5%, dominated by χ_{top}). The corresponding pre-specified 3σ analysis window is [12.29, 13.43] GHz, resolved in bins of width Δf_a .
- **Illustrative coefficient scenario (c_ν varied over $[1/2, 2]$).** Via Eq. (6) this non-statistical sensitivity scenario broadens the target range to 6.4–25.7 GHz. It is a scan, not a point-target test, and must carry its full trials factor and sensitivity variation across the band.
- **Extreme envelope ($c_\nu \in [1/3, 3]$).** $f_\theta \in [4.3, 38.6]$ GHz (~ 9 times the benchmark band): the full scan cost if c_ν is never fixed by theory.

Signal power and sensitivity across the band. Along the curve (6), *at fixed coherent fraction*, the signal powers scale simply: $P_{\text{sig}}^{\text{stack}} \propto g^2/m_\theta^2 \propto c_\nu^0$ for the dielectric stack (exactly invariant under c_ν rescaling) and $P_{\text{sig}}^{\text{cavity}} \propto g^2/m_\theta \propto c_\nu^{-1}$ for the cavity, so the integration times of Tables 4–5 apply unchanged to the stack and rescale as c_ν^2 for the cavity at any other benchmark value of c_ν . If instead the misalignment estimate of Sec. 2.6 is imposed, the coherent fraction itself scales as $f_{\text{coh}} \propto f_a^{7/6} \propto c_\nu^{7/6}$, which multiplies both powers accordingly. Across coefficient scenarios the required sensitivity is therefore not constant: at fixed E/N , $g_{\theta\gamma\gamma} \propto c_\nu^{-1}$, while the assumed misalignment abundance also changes with $f_a \propto c_\nu$; consequently, a broad- c_ν program requires a frequency-dependent sensitivity forecast rather than a uniform scan-time rescaling. The $\pm 0.5\%$ window of the v1 draft covered only m_3 and χ_{top} and is superseded.

Trials factor and global significance. The benchmark prediction is a *conditional narrow target* in model-parameter space, but the experimentally searched frequency interval still contains many statistically independent axion-line templates: the 1.14 GHz benchmark window holds $N_{\text{eff}} \simeq 8.9 \times 10^4$ independent bins of width $\Delta f_a = 12.8$ kHz. A local 5σ excess scanned over

the whole window corresponds to only $\sim 2\sigma$ globally, so the discovery statistic must report both a local significance and a global significance corrected for the pre-specified N_{eff} . Pre-registration fixes the analysis window and procedure; it does not remove the trials correction.

Discovery criterion. A candidate must exceed a pre-specified local threshold corresponding to a *global* 5σ false-alarm probability after correction for N_{eff} templates in the benchmark window. For the conservative independent-bin estimate above, $N_{\text{eff}} \simeq 8.9 \times 10^4$ implies a local threshold of approximately 6.9σ . The final N_{eff} shall be obtained from the matched-filter bank and verified with noise-only Monte Carlo. The candidate must also persist under independent re-taking, scale as B_0^2 when the magnet current is varied, match the expected Maxwell–Boltzmann halo lineshape, and be reproduced at a second apparatus with a different magnet before any claim.

Exclusion criterion. Exclusion is reported at two levels. (1) A null result at KSVZ-level sensitivity, $g_{\theta\gamma\gamma} \leq 2.1 \times 10^{-14} \text{ GeV}^{-1}$ across the preregistered window, excludes the *KSVZ-benchmark realization* of the θ -seesaw target at $f_{\text{coh}} = 0.24$ — and only that realization, since E/N is UV-dependent. (2) A null at DFSZ level ($8.1 \times 10^{-15} \text{ GeV}^{-1}$) additionally covers the DFSZ benchmark; excluding the neutrino–axion linkage itself would further require a fixed E/N assignment and a cosmological determination of f_{coh} . Because exclusion curves assume the full local density, the effective bound on g for a coherent fraction f_{coh} is weaker by $1/\sqrt{f_{\text{coh}}} \approx 2.0$; this rescaling must be applied when interpreting any null.

Side-channel cross-checks. The same f_a predicts $\sum m_\nu = 59 \text{ meV}$ (CMB+BAO, ongoing) and, in the general case, a $0\nu\beta\beta$ window $m_{\beta\beta} \in [0, 6.4] \text{ meV}$, narrowing to the nonzero range $[1.5, 3.7] \text{ meV}$ under the minimal normal-ordering benchmark of Sec. 4.2 (LEGEND-1000, nEXO): these provide framework-level corroboration independent of haloscopes.

7 Honest Caveats

1. **Framework hypothesis and its coefficient.** The θ -seesaw, Eq. (3), is a postulate of the NVG construction. The coefficient c_ν — encoding the two-loop anomaly diagram, possible flavor factors and RG running — is unknown and is propagated as a parameter throughout; $c_\nu = 1$ is a benchmark, not a derived value. Appendix 8 gives a perturbative UV consistency example — a minimal seesaw with a PQ-carrying state N_3 generating m_3 and a PQ-neutral state N_2 generating m_2 — and shows that the benchmark corresponds to a perturbative, τ -Yukawa-scale neutrino coupling with ~ 5 – 10% RG running. It does *not* derive the two-loop QCD-anomaly suppression of Eq. (3), which remains part of the effective hypothesis, and it does not fix c_ν . If Eq. (4) fails — e.g. m_3 from a future $0\nu\beta\beta$ /oscillation combination disagrees with 50.3 meV — the entire frequency target moves. The haloscope test is precisely the discriminator.
2. **Spectral premise.** The input $m_3 = 50.3 \text{ meV}$ assumes normal ordering and a near-minimal lightest neutrino mass. Oscillation experiments measure only mass-squared differences, and cosmology bounds the sum of masses without fixing m_3 at this precision. The target is therefore conditional on this premise; the Monte Carlo window of Sec. 6.4 covers the stated uncertainty, and a shift to inverted ordering or a heavier lightest state would move the target outside the benchmark window.
3. **Model dependence of the coupling.** The factor $|E/N - 1.92|$ in Eq. (9) is benchmarked to KSVZ/DFSZ; a different UV completion rescales $g_{\theta\gamma\gamma}$ without moving the frequency. The mass target is unaffected; the coupling is not predicted.

4. **Coherent fraction.** All powers use $f_{\text{coh}} = 0.24$ (misalignment only). If most of the sector’s density sits in incoherent defects, haloscope signals weaken accordingly; if defects decay into coherent modes, they strengthen. This is the dominant astrophysical uncertainty and is why the exclusion criterion carries the $1/\sqrt{f_{\text{coh}}}$ rescaling.
5. **Technology risk.** $Q_L = 10^6$ at 13 GHz and $\beta^2 = 10^5$ are program goals, not demonstrated hardware (the MADMAX prototype reached $\beta^2 \sim 2000$). The baseline rows ($Q_L = 10^5$, $\beta^2 = 10^4$) reflect the validated state of the art.

8 Conclusion

This note is a phenomenological target paper. Under the stated θ -seesaw hypothesis with benchmark coefficient $c_\nu = 1$, a single neutrino-sector input of the NVG/VMF framework fixes $f_a = 1.07 \times 10^{11}$ GeV, which through lattice-QCD physics yields a conditional point target: $m_\theta = 53.2 \mu\text{eV}$, a microwave line at 12.86 GHz with KSVZ-level photon coupling. The point lands on the QCD axion band within 2.6% (a normalization cross-check, not an independent test), sits in the currently unexplored gap between published ORGAN windows, and is directly on that collaboration’s declared SMASH roadmap. Because the target is a point, the search needs no scanning: a fixed dielectric booster stack reaches discovery-level significance in under a day of integration at design power boost $\beta^2 = 10^4$. The protocol above is preregistered, the frequency window is derived by Monte Carlo from all stated uncertainties, and the exclusion statements are scoped precisely: a KSVZ-level null closes the KSVZ-benchmark realization of the target, not every conceivable θ -model. Either outcome is informative — which is the property a target must have to deserve an experiment.

Appendix A: Reproducibility

All numbers in this note are produced by scripts in the accompanying repository (<https://github.com/infosave2007/vmf>, Zenodo-archived releases):

- `generator_project/calc_theta_haloscope_design.py` (v4) — geometry, signal powers (Eqs. (15)–(16), MADMAX β^2 convention), ORGAN calibration, integration times, published-search status, the c_ν -parametrized θ -seesaw chain, the Monte Carlo block that produces the preregistered window of Sec. 6.4, the trials-factor estimate of the discovery criterion, and the multi-channel interaction block (nucleon/electron/neutrino couplings, minimal normal-ordering flavor benchmark, UV matching) of Sec. 4 and Appendix 8;
- `generator_project/calc_theta_haloscope_v2.py` — frequency derivation, axion-band consistency check (2.6%);
- `verification/nvg_neutrino_seesaw.py` — Eq. (4), fixing f_a ; `verification/nvg_theta_fifth_force.py` — pseudoscalar null channel.

Appendix B: A Perturbative UV Consistency Example for the θ -Seesaw

This appendix shows that the benchmark coefficient $c_\nu = 1$ is *perturbatively viable* — realizable by non-hierarchical UV physics — and quantifies the RG running absorbed into c_ν . Its status is that of a consistency example: it does *not* derive the two-loop QCD-anomaly suppression of Eq. (3), which remains part of the effective NVG hypothesis encoded in c_ν , and it does not fix c_ν .

Field content and PQ charges. A global $U(1)_{\text{PQ}}$ is spontaneously broken by a scalar σ (charge $X_\sigma = 2$) at the scale f_a ; the phase of σ is θ . Two heavy right-handed neutrinos are introduced, as required by the two measured mass splittings (Sec. 4.2):

- N_3 ($X_{N_3} = 1$), with Majorana mass from $\frac{1}{2}y_\sigma\sigma^\dagger N_3^T C N_3$ (PQ-invariant: $-2 + 1 + 1 = 0$): $M_3 = y_\sigma f_a / \sqrt{2} \equiv \kappa f_a$, carrying the PQ phase $M_3(\theta) = M_3 e^{-i\theta/f_a}$. Integrating N_3 out generates the dominant atmospheric mass m_3 and hence fixes f_a ;
- N_2 , PQ-neutral, with an ordinary Majorana mass M_2 and Yukawa $y_2 \bar{L} \tilde{H} N_2$: it generates the solar mass m_2 without a θ phase, so the axion–neutrino coupling resides in the N_3 sector only;
- lepton doublets and charged-lepton singlets with $X_L = X_{e_R} = 1$, so that both $\bar{L} \tilde{H} N_3$ and $\bar{L} H e_R$ are PQ-invariant while quarks and H are neutral;
- one vectorlike color-triplet Q (KSVZ-like, electromagnetically neutral), which makes $U(1)_{\text{PQ}}$ anomalous under color and yields $E/N = 0$ — the photon benchmark of Sec. 2.

Generation of the Weinberg operator. Integrating out N_3 at tree level through the standard seesaw diagram $L \tilde{H} - N_3 - \tilde{H}^\dagger L^\dagger$ produces

$$\mathcal{L}_W = \frac{y_3^2}{2M_3} \frac{\sigma}{|\sigma|} (\tilde{H}^\dagger L)^T C (\tilde{H}^\dagger L) + \text{h.c.}, \quad (19)$$

i.e. the Weinberg operator of Eq. (3) with its PQ phase $e^{i\theta/f_a}$ generated automatically (convention-consistent with the sign choice of Eq. (4)). Matching to the normalization of Eq. (4) gives

$$c_\nu = \frac{y_3^2}{2\kappa (\alpha_s/4\pi)^2} = 5.6 \times 10^3 \frac{y_3^2}{\kappa}. \quad (20)$$

What the matching does — and does not — imply. The benchmark $c_\nu = 1$ with $\kappa = 1$ corresponds to $y_3 = 1.3 \times 10^{-2} \approx 1.8 y_\tau$: a perturbative Yukawa of τ -lepton magnitude, with no hierarchy and no fine-tuning. The scenario range $c_\nu \in [1/3, 3]$ corresponds to $y_3 \in [7.7 \times 10^{-3}, 2.3 \times 10^{-2}]/\sqrt{\kappa}$, i.e. a factor of ~ 2 around y_τ . Perturbativity ($y_3^2 \lesssim 4\pi$) alone bounds $c_\nu \lesssim 7 \times 10^4/\kappa$, which is not restrictive: absent additional flavor dynamics, c_ν is not calculable, and the haloscope test of Sec. 6 remains the discriminator. Crucially, the minimal seesaw completion demonstrates only that the benchmark normalization is perturbative; it does *not* derive the proposed QCD-anomaly suppression $(\alpha_s/4\pi)^2$ — in a plain tree-level seesaw that factor does not arise, and it remains part of the effective NVG hypothesis encoded in c_ν . The colored multiplet Q generates the PQ–QCD anomaly of the θ mode but is not connected to the L – H – N_3 diagram above.

RG running. The one-loop running of the Weinberg coefficient in the Standard Model [28],

$$16\pi^2 \frac{dC_W}{d \ln \mu} = (-3g_2^2 + 6y_t^2 + \lambda)C_W + (Y_e^\dagger Y_e)C_W + C_W(Y_e^\dagger Y_e)^T, \quad (21)$$

shifts the coefficient by ~ 5 –10% between $M_3 = f_a$ and the electroweak scale; this is absorbed into c_ν and is subleading relative to the benchmark scenario granularity.

Neutrino coupling and domain walls. The phase $e^{i\theta/f_a}$ in the m_3 term yields the axial coupling $C_{\nu_3} = 1$ ($g_{\theta\nu_3\nu_3} \simeq m_3/f_a$) used in Sec. 4 — the NVG-specific signature absent in plain KSVZ/DFSZ, carried by the N_3 sector alone. The colored sector carries unit anomaly, so the domain-wall number is $N_{\text{DW}} = 1$ and any walls formed in the post-inflationary scenario are unstable, consistent with the cosmology of Sec. 2.6.

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